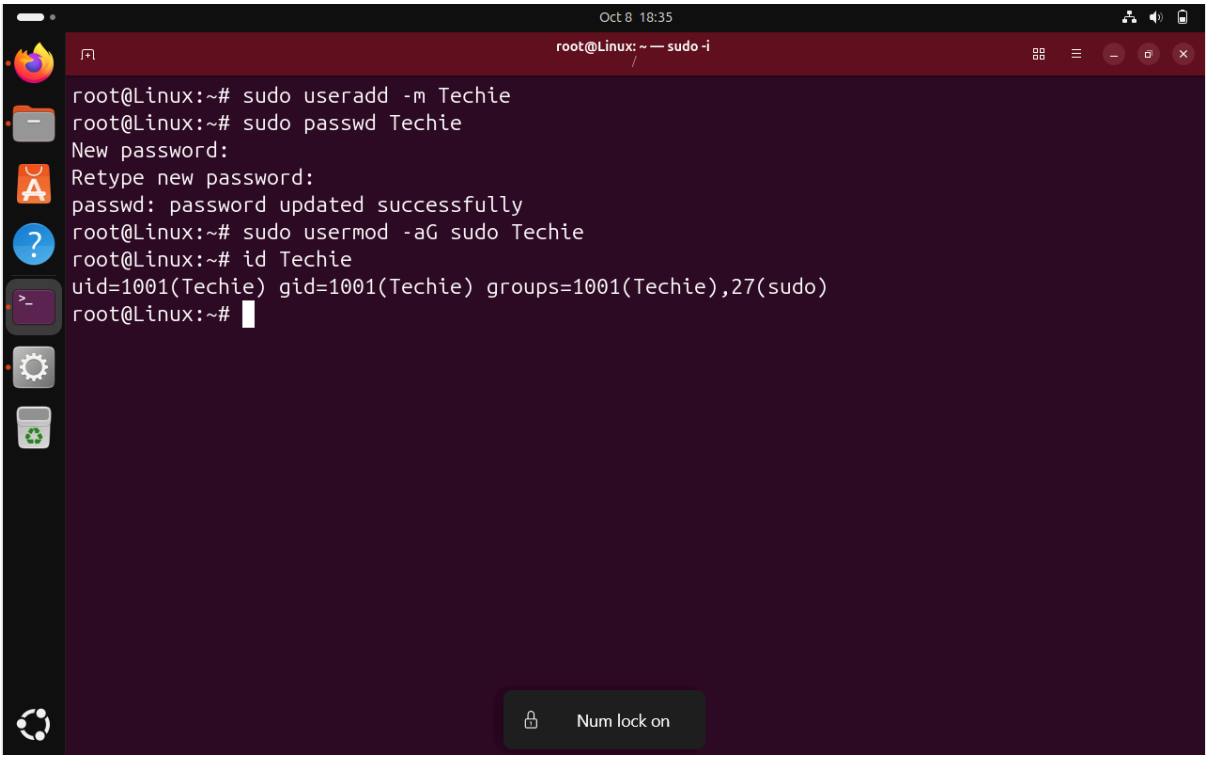
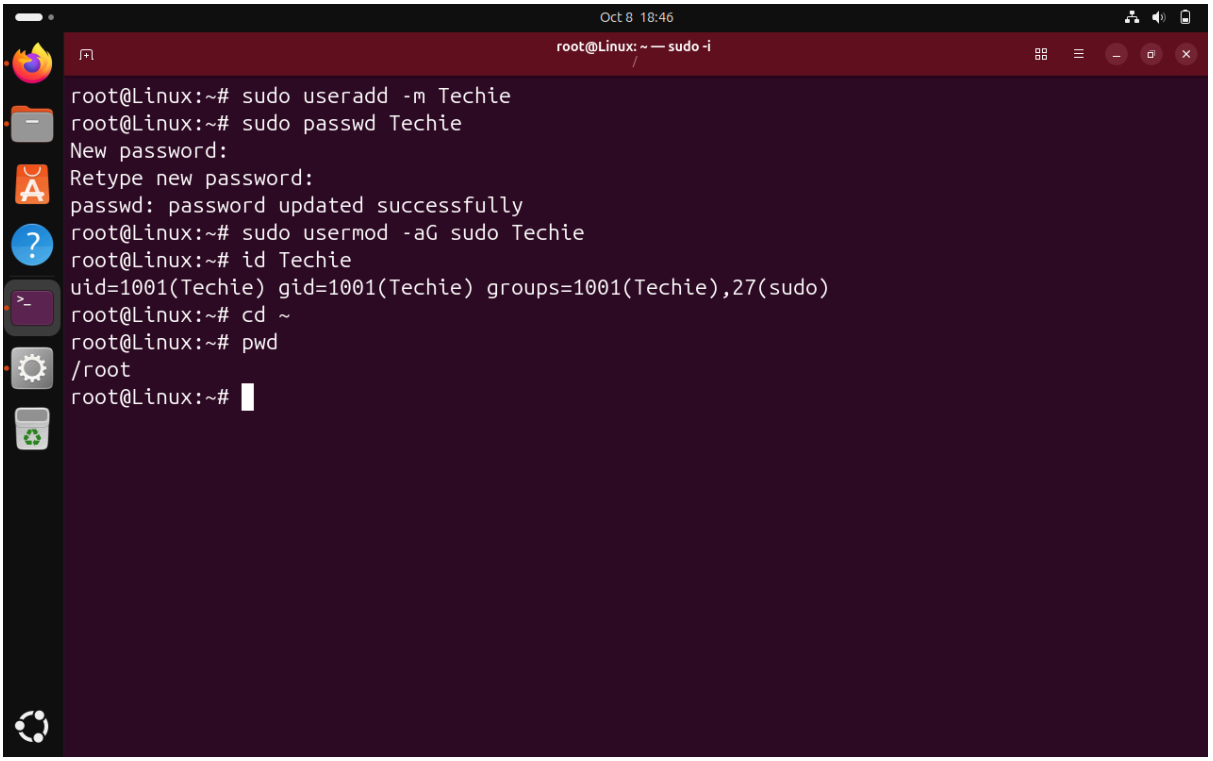
1. Create user Techie and provide sudo access



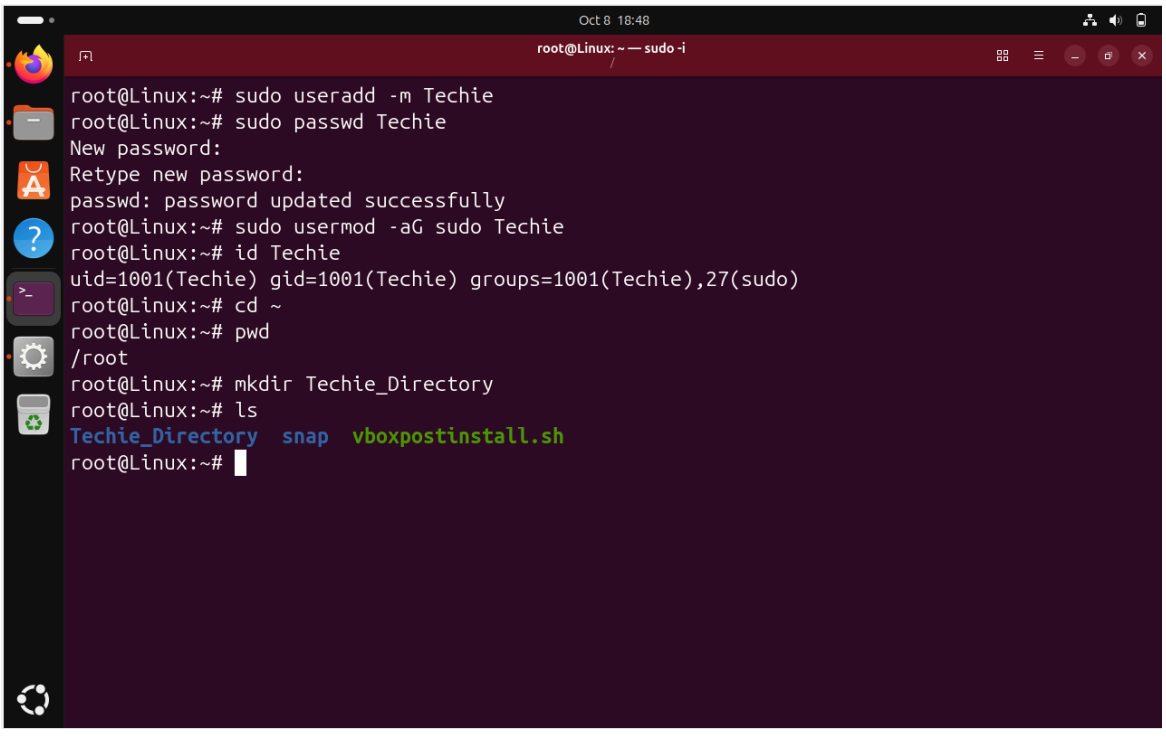
I created a new user named Techie and assigned a password. I added the user to the sudo group so that Techie can perform administrative operations.

2. Navigate to the home directory



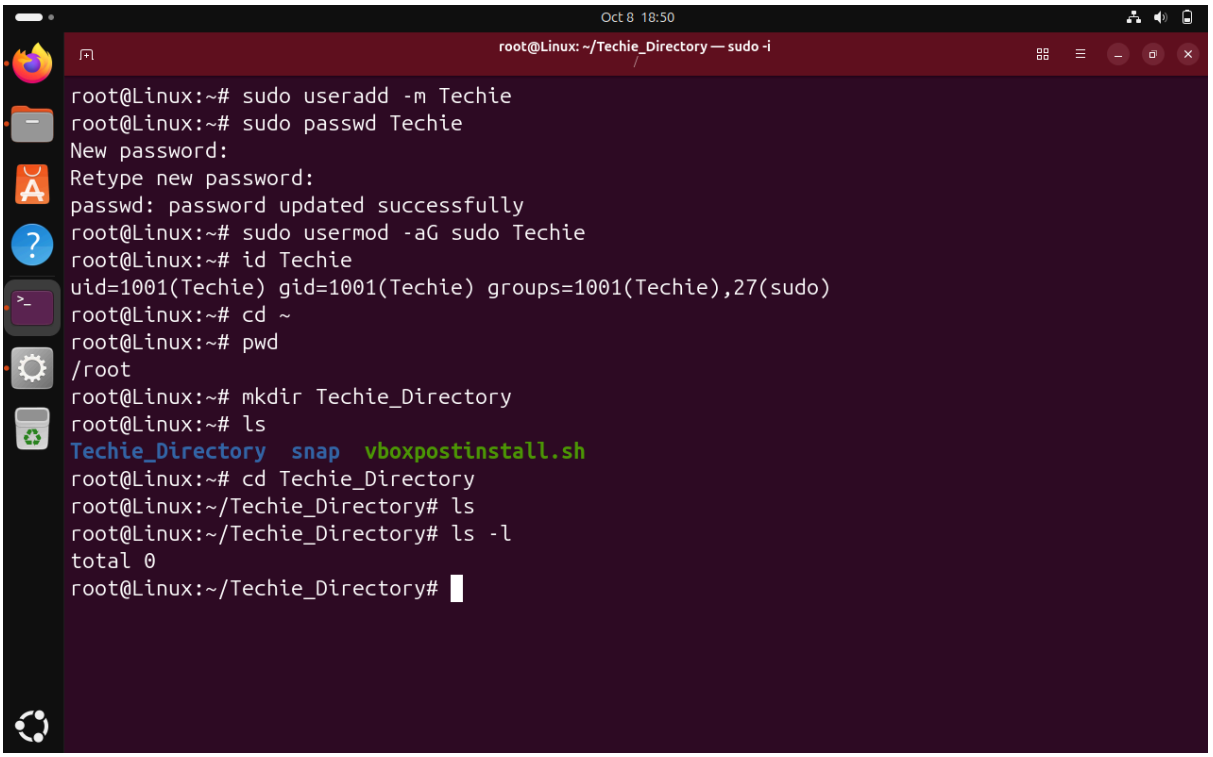
I used the cd command to navigate to the home directory of the current user.

3. Create a new directory.



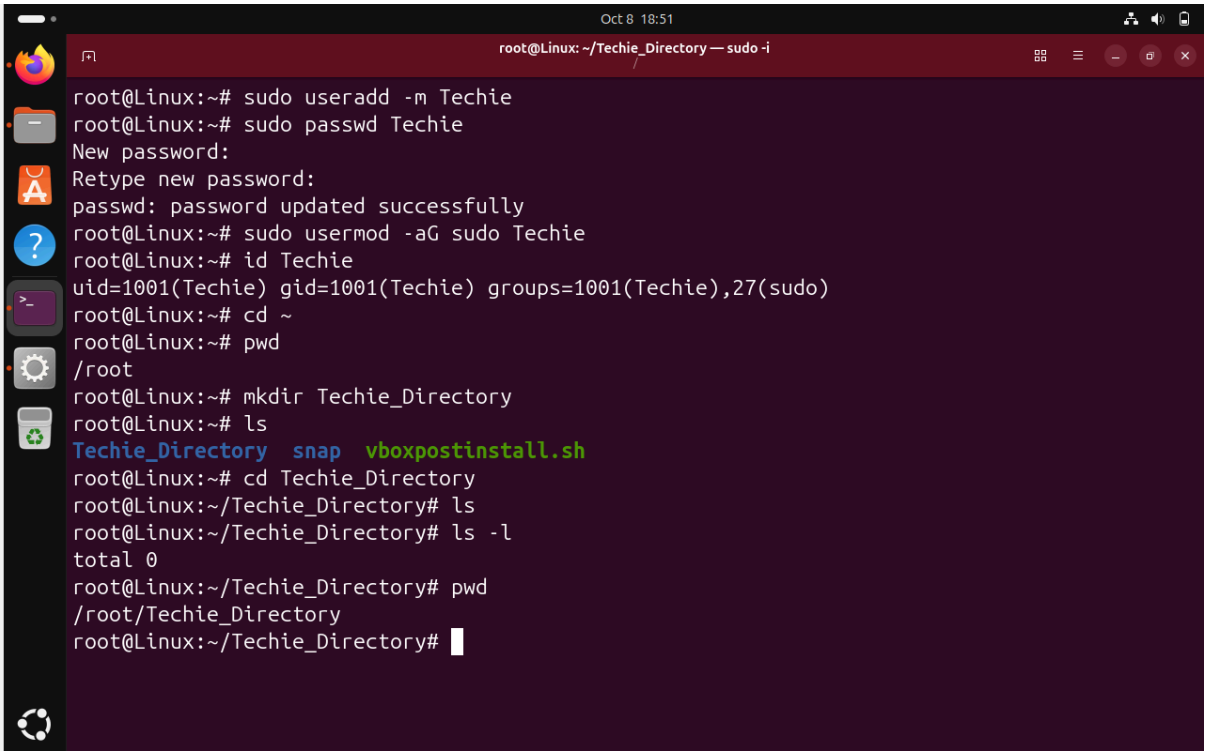
I used the mkdir command to create a new directory named Techie_Directory

4. List the contents of a directory.



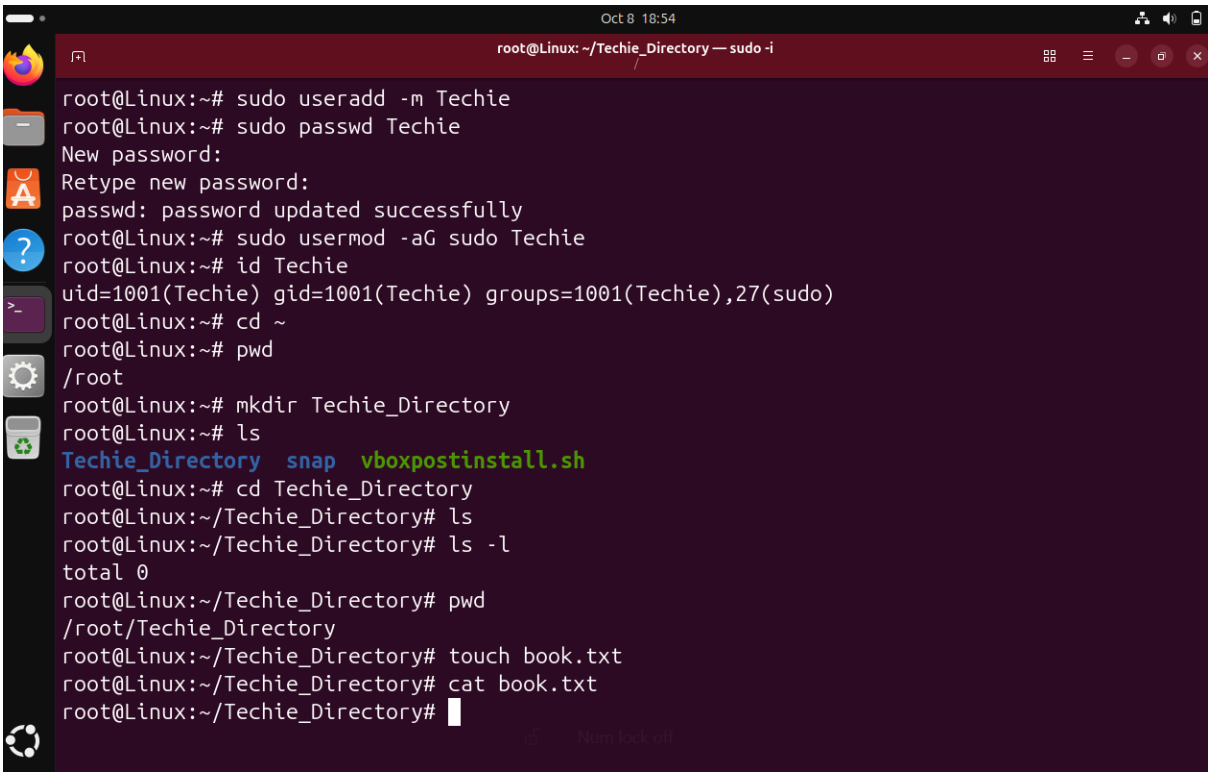
I used the ls command to list the files and directories available in the current directory.

5. Change the current directory



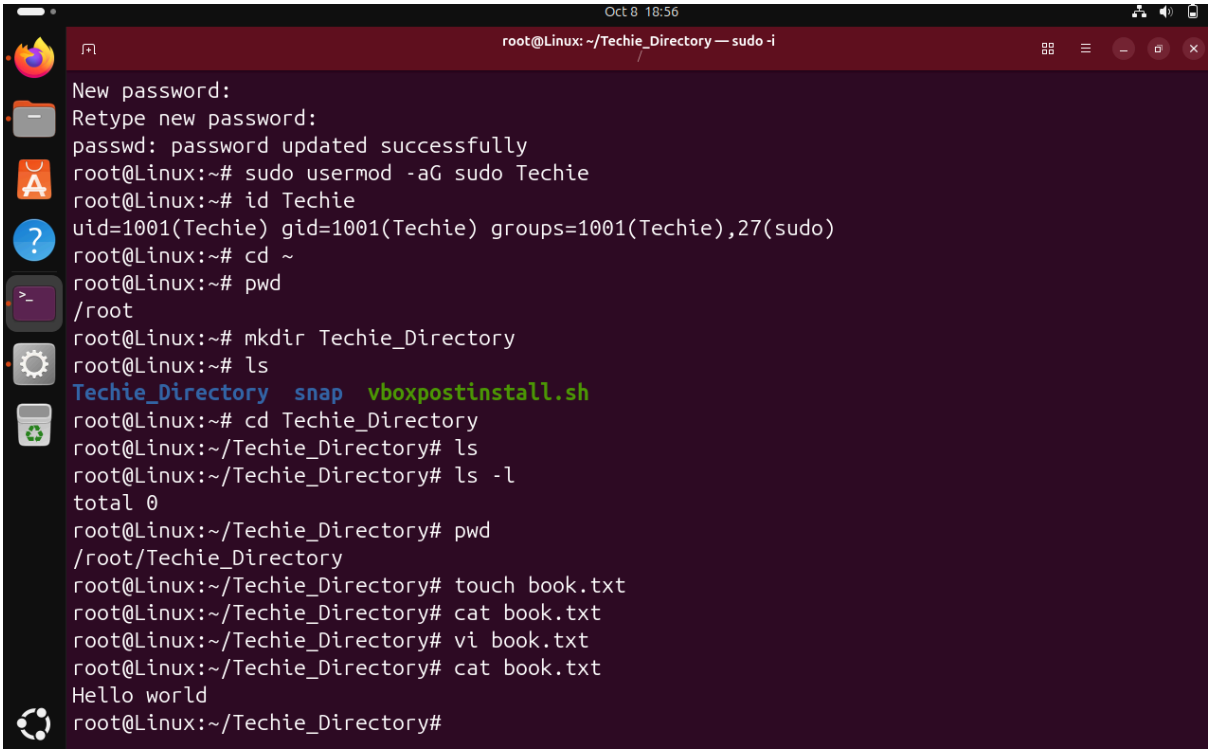
I used the cd command to change from the current directory to Techie_Directory.

6. Create a new empty file



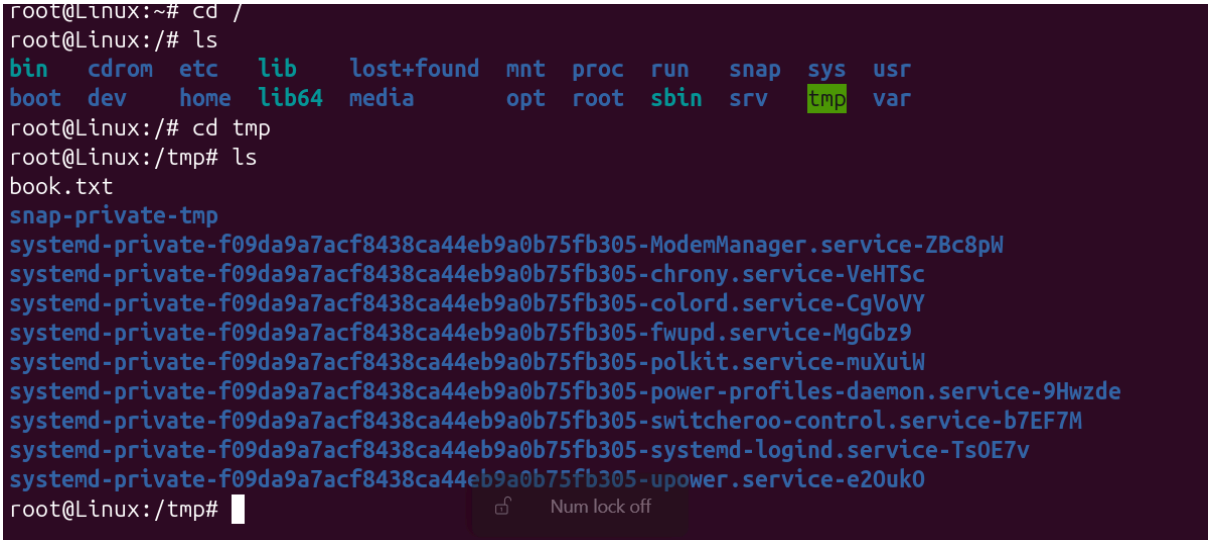
I used the touch command to create a new empty file named book.txt.

7. View the contents of a file



I used the cat command to display the contents of book.txt on the terminal.

8. Copy a file to another location



I used the cp command to create a copy of book.txt and place it in the /tmp directory.

9. Move a file to another location

```
vboxuser@Linux:~$ ls
Desktop    Downloads  Pictures  Templates  book  cloud_training  dog    pen
Documents  Music      Public    Videos     cat   cow             media  snap
```

mv test.txt /home/

I used the mv command to move book.txt from the current directory to the /home directory.

10. Rename a file

```
vboxuser@Linux:~$ touch test
vboxuser@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  book  cat  cloud_training  cow  dog  media  pen  snap  test
vboxuser@Linux:~$ pwd
/home/vboxuser
vboxuser@Linux:~$ mv test student
vboxuser@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  book  cat  cloud_training  cow  dog  media  pen  snap  student
vboxuser@Linux:~$
```

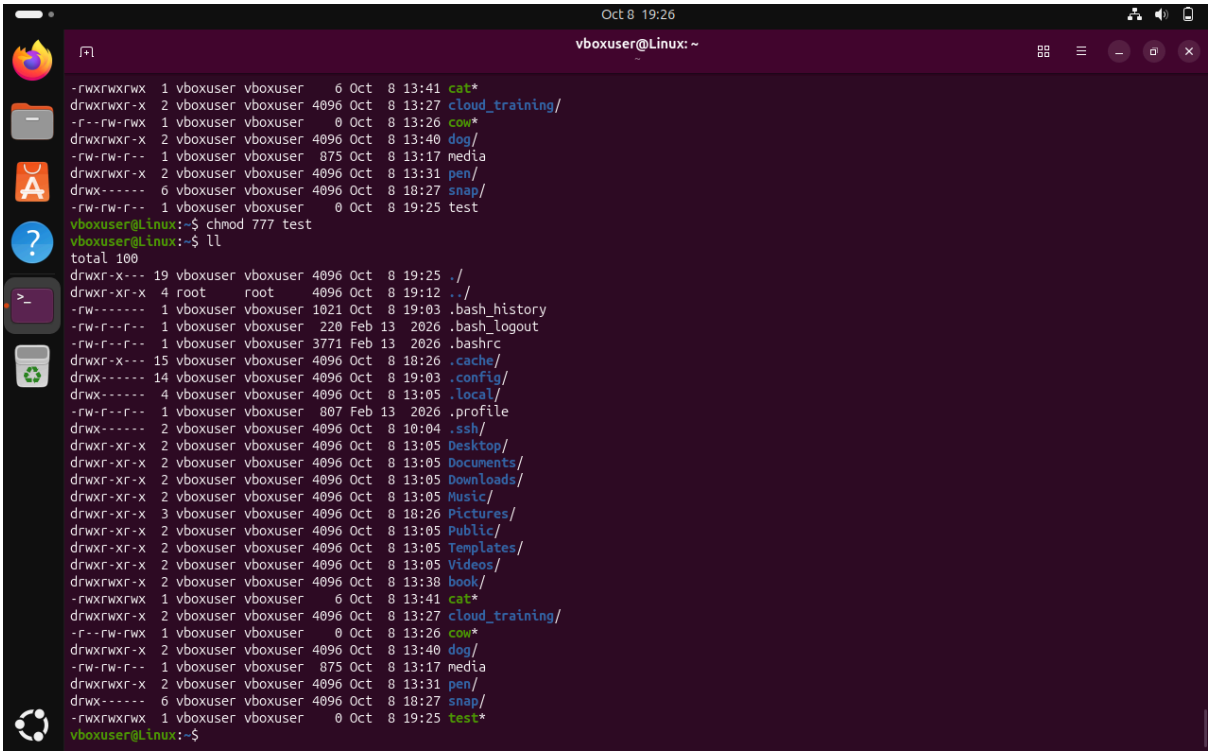
I used the mv command to rename test to student.

11. Delete a file

```
vboxuser@Linux:~$ rm student
vboxuser@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  book  cat  cloud_training  cow  dog  media  pen  snap
vboxuser@Linux:~$
```

I used the rm command to permanently delete the specified file.

12. Grant or revoke permissions on a file or directory



I used the chmod command to modify the permissions of a file or directory. It can be used to grant or remove read, write, and execute permissions.

13. View the current date and time



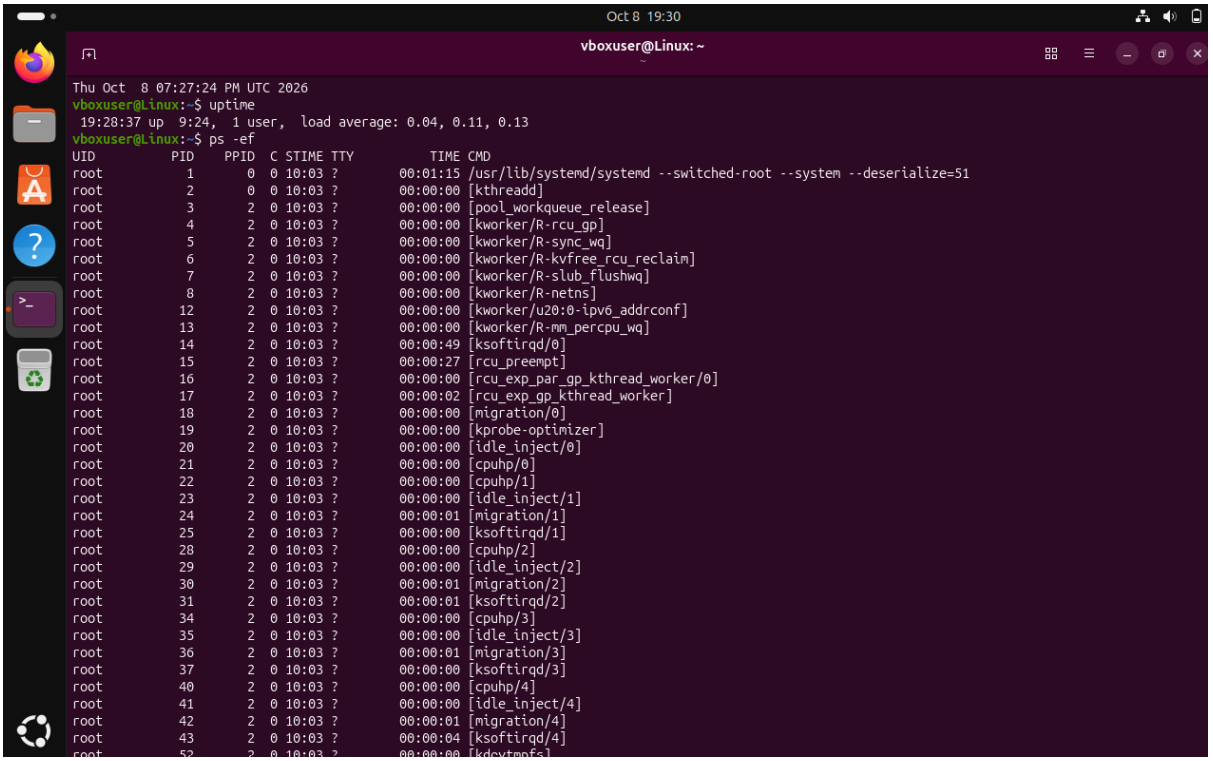
I used the date command to display the current system date, time, and timezone.

14. Check system uptime



I used the uptime command to check how long the system has been running since the last boot.

15. View running processes



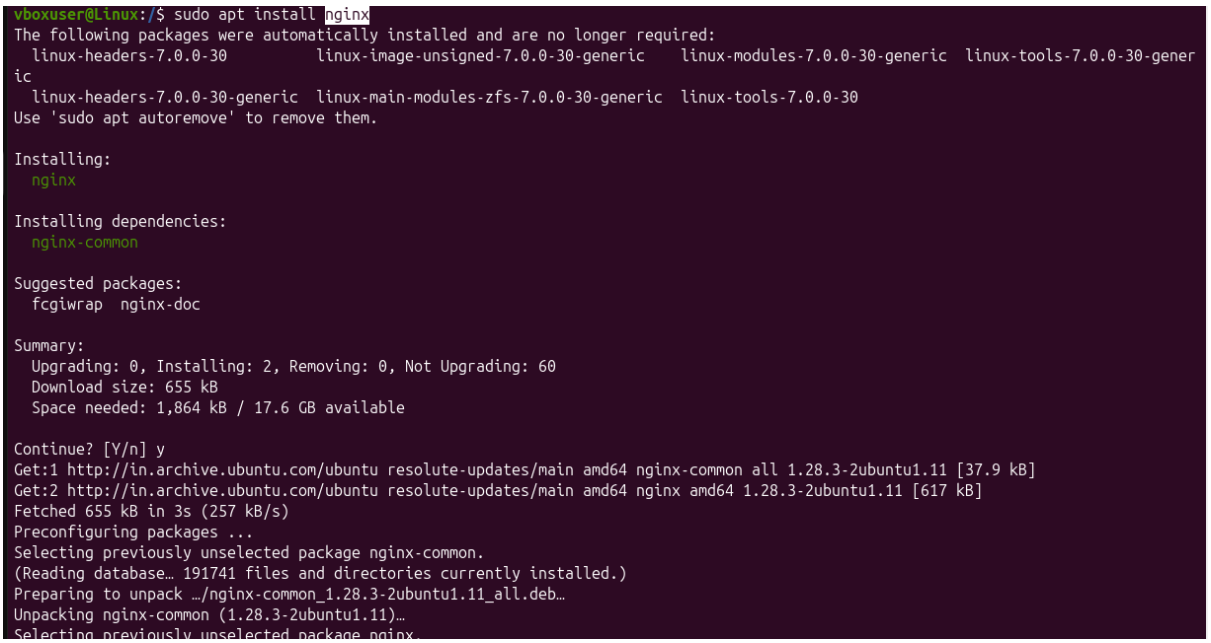
I used the ps -ef command to display the currently running processes along with their process IDs and other details.

16. Kill a running process

```
vboxuser@Linux:~$ sleep 500 &
[1] 20340
vboxuser@Linux:~$ kill 20340
[1]+  Terminated              sleep 500
vboxuser@Linux:~$
```

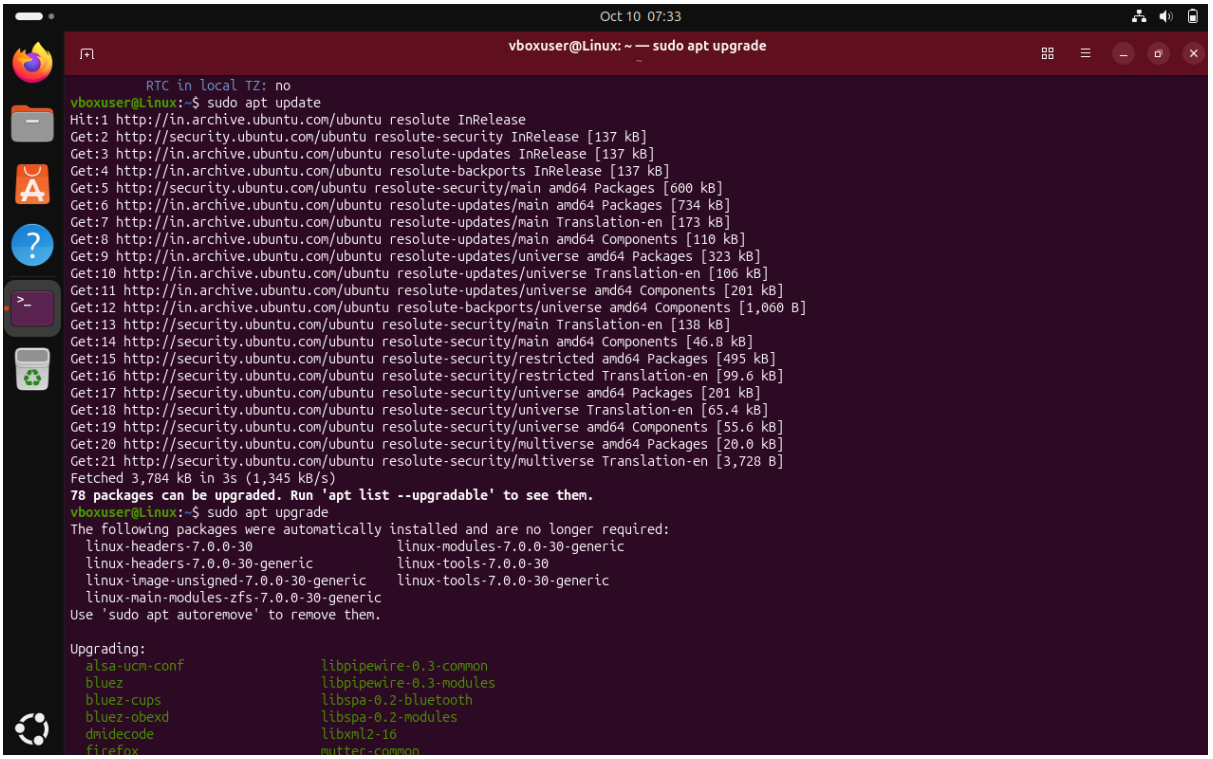
I identified the Process ID (PID) and used the kill command to terminate the process.

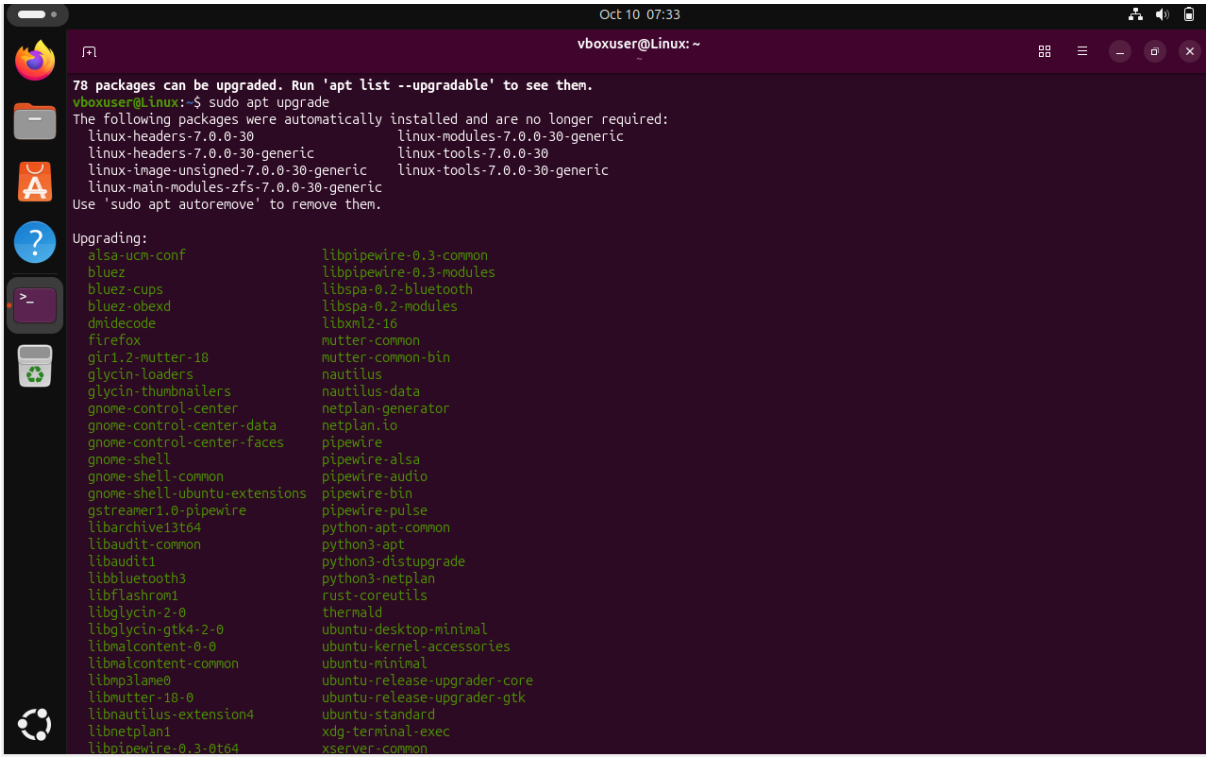
17. Install a package using the package manager



I used the apt package manager to install the Nginx package on the Ubuntu system.

18. Update system packages





I used apt update to refresh the package information and apt upgrade to install available package updates.

19. Create a symbolic link

```
vboxuser@Linux:~$ ln -s /home/vboxuser/file /home/vboxuser/file_link
vboxuser@Linux:~$ ls -l
total 64
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  8 13:05 Desktop
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  8 13:05 Documents
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  8 13:05 Downloads
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  8 13:05 Music
drwxr-xr-x 3 vboxuser vboxuser 4096 Oct  8 18:26 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  8 13:05 Public
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  8 13:05 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  8 13:05 Videos
drwxrwxr-x 2 vboxuser vboxuser 4096 Oct 10 08:01 book
-rwxrwxrwx 1 vboxuser vboxuser    6 Oct  8 13:41 cat
drwxrwxr-x 2 vboxuser vboxuser 4096 Oct  8 13:27 cloud_training
-r--rw-rwx 1 vboxuser vboxuser    0 Oct  8 13:26 cow
drwxrwxr-x 2 vboxuser vboxuser 4096 Oct  8 13:40 dog
-rw-rw-r-- 1 vboxuser vboxuser    6 Oct 10 09:23 file
lrwxrwxrwx 1 vboxuser vboxuser   19 Oct 10 09:25 file_link -> /home/vboxuser/file
-rw-rw-r-- 1 vboxuser vboxuser  875 Oct  8 13:17 media
drwxrwxr-x 2 vboxuser vboxuser 4096 Oct  8 13:31 pen
drwx------ 6 vboxuser vboxuser 4096 Oct  8 18:27 snap
-rwxrwxrwx 1 vboxuser vboxuser    0 Oct  8 19:25 test
vboxuser@Linux:~$ cat file_link
hello
vboxuser@Linux:~$
```

I used ln -s to create a symbolic link pointing to the original file. The symbolic link acts as a reference to the original file.

20. Search for files using find

```
find: ‘/home/Techie’: Permission denied
/home/vboxuser/file
vboxuser@Linux:~$ sudo find /home -name "file"
[sudo: authenticate] Password:
/home/vboxuser/file
vboxuser@Linux:~$
```

I used the find command to search for a file named test.txt inside the /home directory.

21. Compress and decompress files using tar

```
book_link  notes  notes_link
vboxuser@Linux:~/book$ tar -cvf backup.tar notes
notes
vboxuser@Linux:~/book$ ls -l backup.tar
-rw-rw-r-- 1 vboxuser vboxuser 10240 Oct 10 06:04 backup.tar
vboxuser@Linux:~/book$
```

I used tar -cvf to create an archive containing the file. I used tar -xvf to extract the files from the archive.

```
vboxuser@Linux:~/book$ tar -xvf backup.tar
notes
vboxuser@Linux:~/book$
```

22. Monitor system resources

```
vboxuser@Linux:~/book$ top

top - 06:11:15 up 11 min,  1 user,  load average: 0.00, 0.06, 0.07
Tasks: 237 total,   1 running, 236 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.5 us,  0.4 sy,  0.0 ni, 97.1 id,  0.0 wa,  0.0 hi,  2.0 si,  0.0 st
MiB Mem :   5622.8 total,   3067.9 free,   1278.6 used,   1538.9 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   4344.2 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   3063 vboxuser  20   0 5534756 367568 168896 S   3.0   6.4   0:14.95 gnome-shell
   4052 vboxuser  20   0 3086936 387600 144204 S   2.7   6.7   0:08.14 ptyxis
     60 root      20   0       0      0      0 S   0.3   0.0   0:01.53 kcompactd0
   3138 vboxuser  20   0    8616   5212   4716 S   0.3   0.1   0:00.05 dbus-daemon
   3487 vboxuser  20   0  392484   9524   8280 S   0.3   0.2   0:00.11 gvfs-afc-volume
      1 root      20   0   25720  16796  11640 S   0.0   0.3   0:02.74 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.02 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-kblockd
     11 root      20   0       0      0      0 I   0.0   0.0   0:00.15 kworker/0:1-events
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker/u20:0-ipv6_addrconf
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.02 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.80 rcu_preempt
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.18 rcu_exp_gp_kthread_worker
     18 root      rt   0       0      0      0 S   0.0   0.0   0:00.12 migration/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kprobe-optimizer
     20 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     21 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     22 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     23 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
     24 root      rt   0       0      0      0 S   0.0   0.0   0:00.12 migration/1
```

I used the top command to monitor CPU usage, memory usage, load average, and running processes in real time.

23. Create and manage user groups

```
vboxuser@Linux:~$ sudo groupadd Techie_horizon
[sudo: authenticate] Password:
vboxuser@Linux:~$ gpasswd Techie_horizon
gpasswd: Permission denied.
vboxuser@Linux:~$ sudo gpasswd Techie_horizon
Changing the password for group Techie_horizon
New Password:
Re-enter new password:
vboxuser@Linux:~$ sudo usermod -aG Techie_horizon Techie
vboxuser@Linux:~$ groups Techie
Techie : Techie sudo Techie_horizon
vboxuser@Linux:~$
```

I created a Techie_horizon group and added the Techie user to it. Groups help manage permissions for multiple users.

24. Set up SSH passwordless authentication

```
vboxuser@Linux:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [734 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [323 kB]
Fetched 1,333 kB in 3s (425 kB/s)
18 packages can be upgraded. Run 'apt list --upgradable' to see them.
vboxuser@Linux:~$ sudo apt install openssh-server -y
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30                linux-modules-7.0.0-30-generic
  linux-headers-7.0.0-30-generic        linux-tools-7.0.0-30
  linux-image-unsigned-7.0.0-30-generic linux-tools-7.0.0-30-generic
  linux-main-modules-zfs-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.

Upgrading:
  openssh-client

Installing:
  openssh-server

Installing dependencies:
  ncurses-term  openssh-sftp-server  ssh-import-id

Suggested packages:
  molly-guard  monkeysphere  ssh-askpass

Summary:
  Upgrading: 1, Installing: 4, Removing: 0, Not Upgrading: 17
  Download size: 1,885 kB
  Space needed: 7,586 kB / 17.5 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 openssh-client amd64 1:10.2p1-2ubuntu3.7 [936 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 ncurses-term all 6.6+20251231-1 [280 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 openssh-sftp-server amd64 1:10.2p1-2ubuntu3.7 [37.4 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 openssh-server amd64 1:10.2p1-2ubuntu3.7 [622 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 ssh-import-id all 5.11-0ubuntu4 [10.2 kB]
```

```
     Memory: 1.2M (peak: 2.2M)
        CPU: 23ms
     CGroup: /system.slice/ssh.service
             └─10535 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Oct 10 08:44:09 Linux systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Oct 10 08:44:09 Linux sshd[10535]: Server listening on 0.0.0.0 port 22.
Oct 10 08:44:09 Linux sshd[10535]: Server listening on :: port 22.
Oct 10 08:44:09 Linux systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
vboxuser@Linux:~$ ssh-copy-id vboxuser@localhost
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is: SHA256:4D+vXnrZ2yri2gERVUaYLoeh+Rnr4kPvwIMVkWWDeEk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
vboxuser@localhost's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'vboxuser@localhost'"
and check to make sure that only the key(s) you wanted were added.

vboxuser@Linux:~$ ssh vboxuser@localhost
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

vboxuser@Linux:~$
```

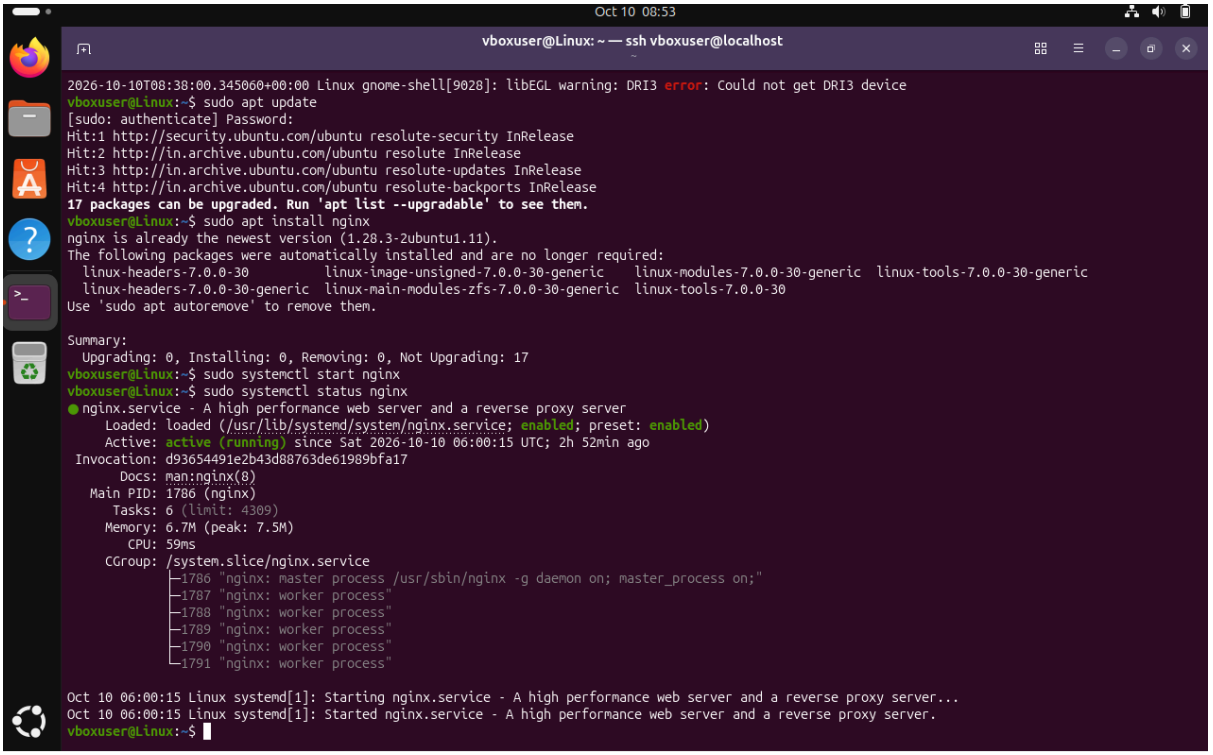
I installed and configured the OpenSSH server on Ubuntu and generated SSH keys for authentication. I copied the public key to the authorized_keys file and successfully tested passwordless SSH login using ssh vboxuser@localhost.

25. Monitor log files using tail or grep

```
vboxuser@Linux:~$ tail -f /var/log/syslog
2026-10-10T08:46:09.492526+00:00 Linux systemd[7510]: Finished update-notifier-release.service - Notification regarding a new release of Ubuntu.
2026-10-10T08:46:12.385131+00:00 Linux systemd[1]: Started session-15.scope - Session 15 of User vboxuser.
2026-10-10T08:46:12.630738+00:00 Linux im-config[10784]: @ IM_CONFIG_ENTRY='profile' XDG_SESSION_TYPE ='tty' UID=1000 PID=10725 PPID=10724
2026-10-10T08:46:12.637881+00:00 Linux im-config[10785]:   GTK_IM_MODULE='' QT_IM_MODULE='' CLUTTER_IM_MODULE='' SDL_IM_MODULE='' XMODIFIERS=''
2026-10-10T08:46:12.645696+00:00 Linux im-config[10788]:   Set IM_CONFIG_CURRENT_DESKTOP <- 'UNDEFINED'
2026-10-10T08:48:39.475900+00:00 Linux PackageKit: daemon quit
2026-10-10T08:48:39.488330+00:00 Linux systemd[1]: packagekit.service: Deactivated successfully.
2026-10-10T08:50:01.818060+00:00 Linux systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
2026-10-10T08:50:01.879180+00:00 Linux systemd[1]: sysstat-collect.service: Deactivated successfully.
2026-10-10T08:50:01.879265+00:00 Linux systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
^C
vboxuser@Linux:~$ grep "error" /var/log/syslog
2026-10-08T10:04:00.612388+00:00 Linux systemd[1]: apport-autoreport.path - Process error reports when automatic reporting is enabled (file watch) ski
pped, unmet condition check ConditionPathExists=/var/lib/apport/autoreport
2026-10-08T10:04:00.612398+00:00 Linux systemd[1]: apport-autoreport.timer - Process error reports when automatic reporting is enabled (timer based) s
kipped, unmet condition check ConditionPathExists=/var/lib/apport/autoreport
2026-10-08T10:04:01.285281+00:00 Linux alsactl[1404]: alsa-lib main.c:1804:(snd_use_case_mgr_open) [error.ucm] failed to import hw:0 use case configur
ation -2
2026-10-08T10:09:02.341324+00:00 Linux kernel: audit: type=1400 audit(1791454142.338:256): apparmor="DENIED" operation="getattr" class="file" info="Fa
iled name lookup - disconnected path" error=-13 profile="unprivileged_userns" name="" pid=3238 comm="systemd-detect-" requested_mask="r" denied_mask="
r" fsuid=60578 ouid=0
2026-10-08T12:43:11.237413+00:00 Linux snapd[3721]: stateengine.go:161: state ensure error: Get "https://api.snapcraft.io/api/v1/snaps/sections": dial
 tcp: lookup api.snapcraft.io on 127.0.0.53:53: read udp 127.0.0.1:36403->127.0.0.53:53: i/o timeout
2026-10-08T13:05:05.804971+00:00 Linux kernel: audit: type=1400 audit(1791464705.803:292): apparmor="DENIED" operation="getattr" class="file" info="Fa
iled name lookup - disconnected path" error=-13 profile="unprivileged_userns" name="" pid=5210 comm="systemd-detect-" requested_mask="r" denied_mask="
r" fsuid=1000 ouid=0
2026-10-08T13:05:06.912887+00:00 Linux ubuntu-insights[5162]: 2026/10/08 13:05:06 WARN failed to collect screen info error="failed to run xrandr: exit
 status 1"
2026-10-08T13:05:08.962340+00:00 Linux gnome-shell[5759]: libinput error: event6  - VirtualBox mouse integration: client bug: event processing lagging
 behind by 21ms, your system is too slow
2026-10-08T13:05:13.675581+00:00 Linux gnome-initial-setup[6133]: MESA: error: ZINK: failed to choose pdev
2026-10-08T13:05:13.780890+00:00 Linux gnome-shell[6726]: libEGL warning: DRI3 error: Could not get DRI3 device
2026-10-08T13:05:27.188653+00:00 Linux gnome-initial-setup[6133]: time=2026-10-08T13:05:27.188Z level=INFO msg="failed to read file" file=/sys/devices
/pci0000:00/0000:00:02.0/label error="open /sys/devices/pci0000:00/0000:00:02.0/label: no such file or directory"
2026-10-08T13:10:50.008571+00:00 Linux xdg-terminal-exec[7595]: MESA: error: ZINK: failed to choose pdev
2026-10-08T15:10:29.511896+00:00 Linux kernel: audit: type=1400 audit(1791472229.506:309): apparmor="DENIED" operation="getattr" class="file" info="Fa
iled name lookup - disconnected path" error=-13 profile="unprivileged_userns" name="" pid=10115 comm="systemd-detect-" requested_mask="r" denied_mask=
"r" fsuid=60579 ouid=0
```

I used tail -f to monitor new log entries in real time. I used grep to search the log file for a specific pattern such as error.

26. Set up a web server – Nginx



I installed and started the Nginx web server. I used systemctl status to verify whether the service is running.

27. Configure and secure MySQL database

```
vboxuser@Linux:~$ sudo apt install mysql-server
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30          linux-image-unsigned-7.0.0-30-generic    linux-modules-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-headers-7.0.0-30-generic  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30
Use 'sudo apt autoremove' to remove them.

Installing:
  mysql-server

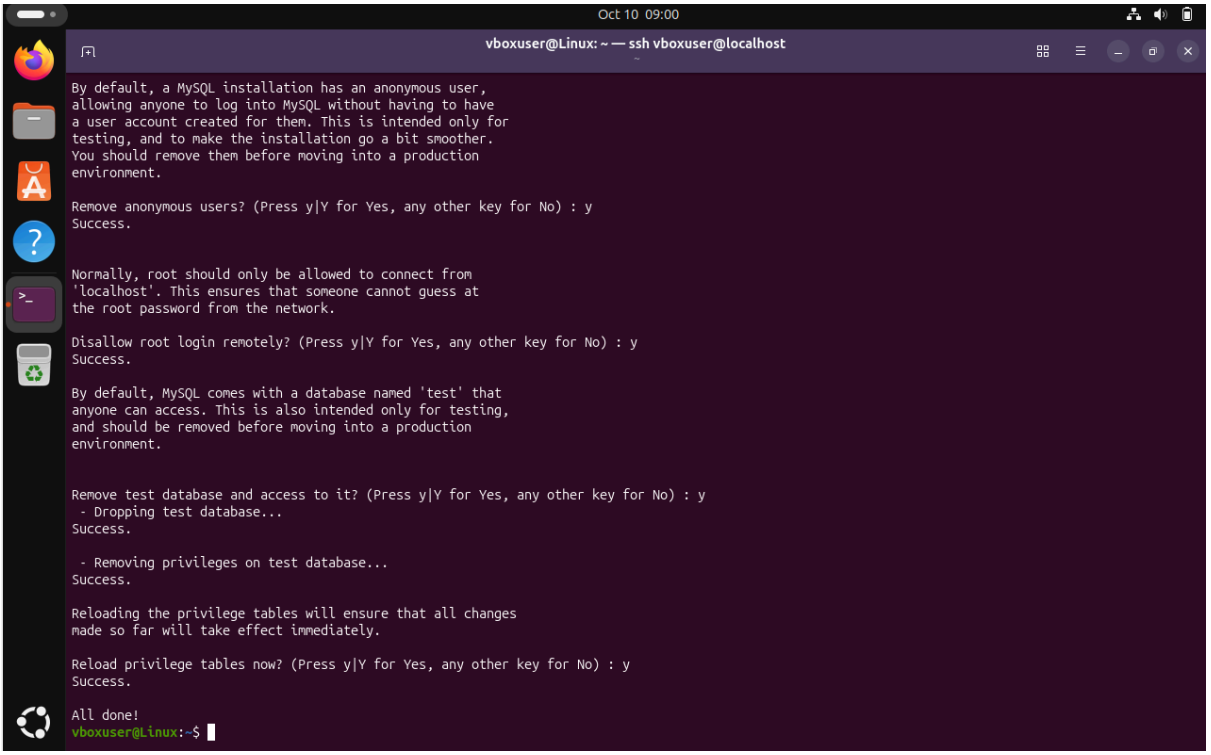
Installing dependencies:
  libaio1t64        libfcgi-bin   libgoogle-perftools4t64  libprotobuf-lite32t64    mysql-client-core
  libcgi-fast-perl  libfcgi-perl  libhtml-template-perl    libtcmalloc-minimal4t64  mysql-common
  libcgi-pm-perl    libfcgi0t64   libmecab2                mysql-client             mysql-server-core

Suggested packages:
  libipc-sharedcache-perl

Recommended packages:
  mecab-ipadic-utf

Summary:
  Upgrading: 0, Installing: 16, Removing: 0, Not Upgrading: 17
  Download size: 24.4 MB
  Space needed: 182 MB / 17.4 GB available

Continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 mysql-common all 5.8+1.1.1ubuntu2 [7,002 B]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libtcmalloc-minimal4t64 amd64 2.18.1-1 [138 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libgoogle-perftools4t64 amd64 2.18.1-1 [199 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 mysql-client-core amd64 8.4.11-0ubuntu0.26.04.1 [2,200 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 mysql-client amd64 8.4.11-0ubuntu0.26.04.1 [9,126 B]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libaio1t64 amd64 0.3.113-8build1 [7,356 B]
Get:7 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libmecab2 amd64 0.996-15.1build4 [205 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libprotobuf-lite32t64 amd64 3.21.12-15ubuntu1 [251 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 mysql-server-core amd64 8.4.11-0ubuntu0.26.04.1 [19.4 MB]
Get:10 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 mysql-server amd64 8.4.11-0ubuntu0.26.04.1 [1,596 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libcgi-pm-perl all 4.71-1build1 [185 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libfcgi0t64 amd64 2.4.5-0.1build2 [27.9 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libfcgi-perl amd64 0.82+ds-3build3 [21.7 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libcgi-fast-perl all 1:2.17-1 [10.3 kB]
```



I installed MySQL and started the database service. I used mysql_secure_installation to perform basic security configuration such as removing unnecessary/default setting

28. Set up an Application Server – Apache Tomcat

```
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
17 packages can be upgraded. Run 'apt list --upgradable' to see them.
vboxuser@Linux:~$ sudo apt install default-jdk
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30         linux-image-unsigned-7.0.0-30-generic    linux-modules-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-headers-7.0.0-30-generic  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30
Use 'sudo apt autoremove' to remove them.

Installing:
  default-jdk

Installing dependencies:
  ca-certificates-java  fonts-dejavu-extra       libice-dev  libxcb1-dev     openjdk-25-jdk-headless  x11proto-dev
  default-jdk-headless  java-common              libsm-dev   libxdmcp-dev    openjdk-25-jre           xorg-sgml-doctools
  default-jre           libatk-wrapper-java      libx11-dev  libxt-dev       openjdk-25-jre-headless  xtrans-dev
  default-jre-headless  libatk-wrapper-java-jni  libxau-dev  openjdk-25-jdk  uuid-dev

Suggested packages:
  libice-doc  libx11-doc  libxt-doc        openjdk-25-source  fonts-ipafont-gothic  fonts-wqy-microhei  fonts-indic
  libsm-doc   libxcb-doc  openjdk-25-demo  visualvm           fonts-ipafont-mincho  | fonts-wqy-zenhei

Summary:
  Upgrading: 0, Installing: 24, Removing: 0, Not Upgrading: 17
  Download size: 155 MB
  Space needed: 364 MB / 17.1 GB available

Continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 ca-certificates-java all 20260311 [11.3 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 java-common all 0.77 [6,682 B]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 openjdk-25-jre-headless amd64 25.0.4.1+1-1~26.04.4 [59.0 MB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 default-jre-headless amd64 2:1.25-77 [2,986 B]
Get:5 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 openjdk-25-jre amd64 25.0.4.1+1-1~26.04.4 [219 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 default-jre amd64 2:1.25-77 [872 B]
Get:7 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 openjdk-25-jdk-headless amd64 25.0.4.1+1-1~26.04.4 [89.4 MB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 default-jdk-headless amd64 2:1.25-77 [916 B]
Get:9 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 openjdk-25-jdk amd64 25.0.4.1+1-1~26.04.4 [1,782 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 default-jdk amd64 2:1.25-77 [882 B]
Get:11 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 fonts-dejavu-extra all 2.37-8build1 [1,944 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libatk-wrapper-java all 0.44.0-1 [55.2 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libatk-wrapper-java-jni amd64 0.44.0-1 [48.0 kB]
```

```
vboxuser@Linux: ~ — ssh vboxuser@localhost

Creating user 'tomcat' (Apache Tomcat) with UID 972 and GID 972.
Creating config file /etc/tomcat10/tomcat-users.xml with new version
Creating config file /etc/tomcat10/web.xml with new version
Creating config file /etc/tomcat10/server.xml with new version
Creating config file /etc/tomcat10/logging.properties with new version
Creating config file /etc/tomcat10/context.xml with new version
Creating config file /etc/tomcat10/catalina.properties with new version
Creating config file /etc/tomcat10/jaspic-providers.xml with new version
Creating config file /etc/logrotate.d/tomcat10 with new version
Creating config file /etc/default/tomcat10 with new version
Created symlink '/etc/systemd/system/multi-user.target.wants/tomcat10.service' → '/usr/lib/systemd/system/tomcat10.service'.
Processing triggers for rsyslog (8.2512.0-1ubuntu4.2)…
Processing triggers for libc-bin (2.43-2ubuntu2.4)…
vboxuser@Linux:~$ sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Server
     Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-10-10 09:18:54 UTC; 30s ago
 Invocation: c95e9230cf7a48b59d3df2b37678b9b5
       Docs: https://tomcat.apache.org/tomcat-10.0-doc/index.html
    Process: 14443 ExecStartPre=/usr/libexec/tomcat10/tomcat-update-policy.sh (code=exited, status=0/SUCCESS)
   Main PID: 14449 (java)
      Tasks: 38 (limit: 4309)
     Memory: 128.8M (peak: 129.6M)
        CPU: 4.838s
     CGroup: /system.slice/tomcat10.service
             └─14449 /usr/lib/jvm/default-java/bin/java -Djava.util.logging.config.file=/var/lib/tomcat10/conf/logging.properties -Djava.util.logging>

Oct 10 09:18:54 Linux tomcat10[14449]: OpenSSL successfully initialized [OpenSSL 3.5.5 27 Jan 2026]
Oct 10 09:18:55 Linux tomcat10[14449]: Initializing ProtocolHandler ["http-nio-8080"]
Oct 10 09:18:55 Linux tomcat10[14449]: Server initialization in [626] milliseconds
Oct 10 09:18:55 Linux tomcat10[14449]: Starting service [Catalina]
Oct 10 09:18:55 Linux tomcat10[14449]: Starting Servlet engine: [Apache Tomcat/10.1.55 (Ubuntu)]
Oct 10 09:18:55 Linux tomcat10[14449]: Deploying web application directory [/var/lib/tomcat10/webapps/ROOT]
Oct 10 09:18:56 Linux tomcat10[14449]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for a comple>
Oct 10 09:18:56 Linux tomcat10[14449]: Deployment of web application directory [/var/lib/tomcat10/webapps/ROOT] has finished in [971] ms
Oct 10 09:18:56 Linux tomcat10[14449]: Starting ProtocolHandler ["http-nio-8080"]
Oct 10 09:18:56 Linux tomcat10[14449]: Server startup in [1139] milliseconds

vboxuser@Linux:~$
```

I installed Java as the required runtime and configured Apache Tomcat as the application server. I started Tomcat and verified that its process was running.

29. Create a service file for Apache Tomcat

```
[Unit]
Description=Apache Tomcat 10 Web Application Server
Documentation=https://tomcat.apache.org/tomcat-10.0-doc/index.html
After=network.target
RequiresMountsFor=/var/log/tomcat10 /var/lib/tomcat10

[Service]

# Configuration
Environment="CATALINA_HOME=/usr/share/tomcat10"
Environment="CATALINA_BASE=/var/lib/tomcat10"
Environment="CATALINA_TMPDIR=/tmp"
Environment="JAVA_OPTS=-Djava.awt.headless=true"

# Lifecycle
Type=simple
ExecStartPre=+/usr/libexec/tomcat10/tomcat-update-policy.sh
ExecStart=/bin/sh /usr/libexec/tomcat10/tomcat-start.sh
SuccessExitStatus=143
Restart=on-abort

# Logging
SyslogIdentifier=tomcat10

# Security
User=tomcat
Group=tomcat
PrivateTmp=yes
AmbientCapabilities=CAP_NET_BIND_SERVICE
NoNewPrivileges=true
CacheDirectory=tomcat10
CacheDirectoryMode=750
ProtectSystem=strict

vboxuser@Linux:~$ sudo systemctl is-enabled tomcat10
enabled
vboxuser@Linux:~$ sudo systemctl is-active tomcat10
active
vboxuser@Linux:~$
```

I created a systemd service file for Apache Tomcat so that Tomcat can be managed using systemctl. I can start, stop, restart, enable, and check the Tomcat service using systemd.

30. Print specific columns from a delimited file

```
vboxuser@Linux:~$ ls
Desktop    Downloads  Pictures  Templates  book  cloud_training  dog   file_link  pen   saam.pub  test
Documents  Music      Public    Videos     cat   cow             file  media      saam  snap
vboxuser@Linux:~$ touch students
vboxuser@Linux:~$ vi students
vboxuser@Linux:~$ vi students
vboxuser@Linux:~$ rm students
vboxuser@Linux:~$ touch students
vboxuser@Linux:~$ vi students
vboxuser@Linux:~$ cat students

1001    Aarav Sharma    20      Computer Science        aarav.sharma@example.com        3.82
1002    Diya Patel      19      Biology diya.patel@example.com  3.65
1003    Rohan Verma     21      Mathematics     rohan.verma@example.com 3.91
1004    Sneha Rao       20      Physics sneha.rao@example.com   3.42

vboxuser@Linux:~$ awk -F',' '{print $2, $4}' students
```

31. Filter and print lines based on a pattern/condition

```
vboxuser@Ubuntu:~/gfg$ awk '{print}' employee.txt
ajay manager account 45000
sunil clerk account 25000
varun manager sales 50000
amit manager account 47000
tarun peon sales 50000
deepak clerk sales 23000
sunit peon sales 13000

vboxuser@Ubuntu:~/gfg$
```

```
vboxuser@Ubuntu:~/gfg$ awk -F' ' '{print $1, $4}' employee.txt
ajay 45000
sunil 25000
varun 50000
amit 47000
tarun 50000
deepak 23000
sunit 13000
```

```
vboxuser@Ubuntu:~/gfg$ awk -f print_salary.awk employee.txt
ajay has salary 45000
sunil has salary 25000
varun has salary 50000
amit has salary 47000
tarun has salary 50000
deepak has salary 23000
sunit has salary 13000
 has salary
vboxuser@Ubuntu:~/gfg$
```

32. Calculate average, sum, or statistics of a column

```
Oct 10 12:46:44 Linux tomcat10[24148]: Deployment of web application directory [/var/lib/tomcat10/webapps/ROOT] has finished in [1,33
Oct 10 12:46:44 Linux tomcat10[24148]: Starting ProtocolHandler ["http-nio-8080"]
Oct 10 12:46:44 Linux tomcat10[24148]: Server startup in [1393] milliseconds
vboxuser@Linux:~/devops$ nano students.txt
vboxuser@Linux:~/devops$ cut -d ',' -f 2 students.txt
Ravi
Sita
Arun
Meena

vboxuser@Linux:~/devops$ grep "Java" students.txt
101,Ravi,Java,85
vboxuser@Linux:~/devops$ awk -F ',' 'NR>1 {sum+=$4; count++}
> ^C
vboxuser@Linux:~/devops$ awk -F',' 'NR>1 {sum+=$4; count++} END {print "Sum:",sum; print "Average:",sum/count}' students.txt
Sum: 263
Average: 65.75
vboxuser@Linux:~/devops$
```

33. Perform string manipulation

```
vboxuser@Linux:~$ ls
Desktop    Downloads  Pictures  Templates  book  cloud_training  dog   file_link  media  saam      snap      task41
Documents  Music      Public    Videos     cat   cow             file  hello      pen    saam.pub  students  test
vboxuser@Linux:~$ vi employees
vboxuser@Linux:~$ ls
Desktop    Downloads  Pictures  Templates  book  cloud_training  dog        file       hello  pen   saam.pub  students  test
Documents  Music      Public    Videos     cat   cow             employees  file_link  media  saam  snap      task41
vboxuser@Linux:~$ awk '{print toupper($0)}' employees
101, JOHN , IT
102 , DOG , CSE
103 , CAT , MECH
vboxuser@Linux:~$ awk '{print tolower($0)}' employees
101, john , it
102 , dog , cse
103 , cat , mech
vboxuser@Linux:~$
```

34. Count occurrences of a specific pattern

```
vboxuser@Linux:~/devops$ awk -F ',' 'NR>1 {sum+=$4; count++}
> ^C
vboxuser@Linux:~/devops$ awk -F',' 'NR>1 {sum+=$4; count++} END {print "Sum:",sum; print "Average:",sum/count}' students.txt
Sum: 263
Average: 65.75
vboxuser@Linux:~/devops$ awk -F',' 'NR>1 {print to upper($2)}' students.txt
awk: line 2: function upper never defined
vboxuser@Linux:~/devops$ awk -F',' 'NR>1 {print toupper($2)}' students.txt
SITA
ARUN
MEENA

vboxuser@Linux:~/devops$ grep -o "Java" students.txt | wc -l
1
vboxuser@Linux:~/devops$
```

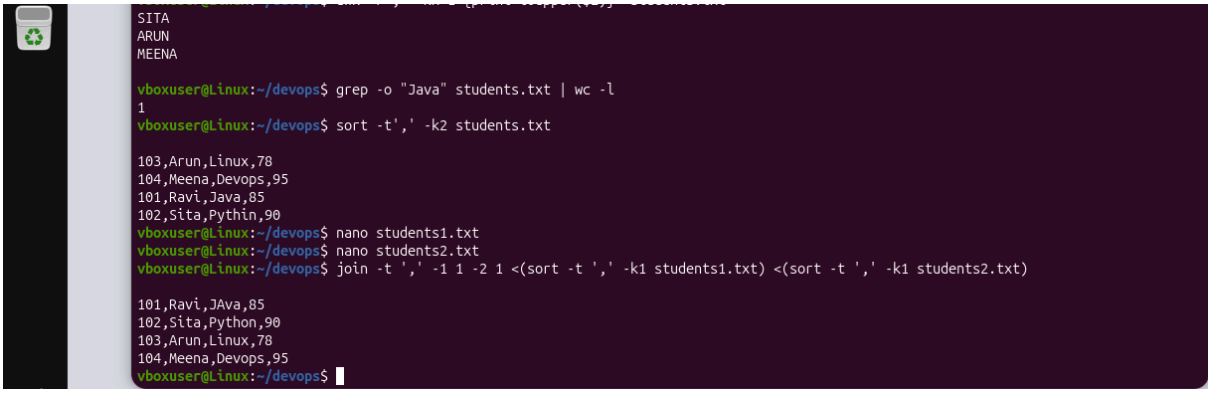
35. Sort lines based on a specific field/column

```
vboxuser@Linux:~/devops$ awk -F',' 'NR>1 {print to upper($2)}' students.txt
awk: line 2: function upper never defined
vboxuser@Linux:~/devops$ awk -F',' 'NR>1 {print toupper($2)}' students.txt
SITA
ARUN
MEENA

vboxuser@Linux:~/devops$ grep -o "Java" students.txt | wc -l
1
vboxuser@Linux:~/devops$ sort -t',' -k2 students.txt

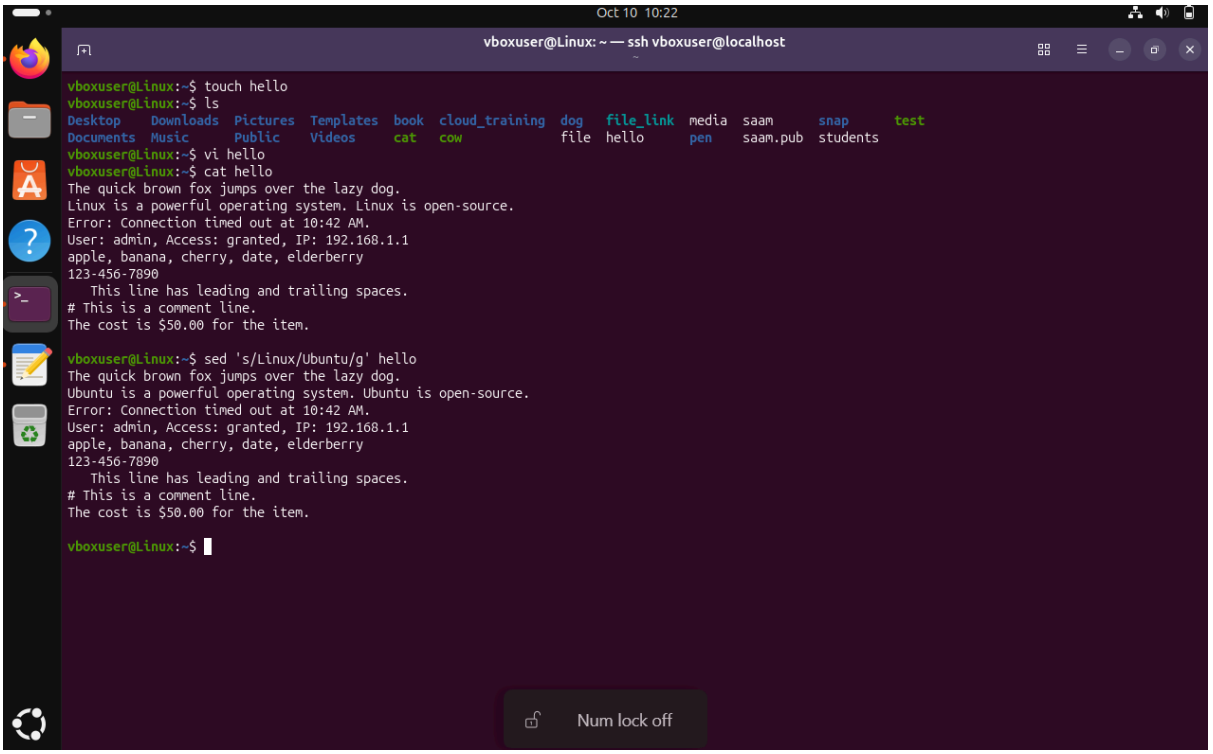
103,Arun,Linux,78
104,Meena,Devops,95
101,Ravi,Java,85
102,Sita,Pythin,90
vboxuser@Linux:~/devops$
```

36. Merge multiple files based on a common field/column



37. Substitute text using search and replace

I used sed to search for the word Linux and replace it with Ubuntu. The g option replaces all occurrences in each line.



38. Delete specific lines based on pattern or line number

```
vboxuser@Linux:~$ sed '/apple/d' hello
The quick brown fox jumps over the lazy dog.
Linux is a powerful operating system. Linux is open-source.
Error: Connection timed out at 10:42 AM.
User: admin, Access: granted, IP: 192.168.1.1
123-456-7890
   This line has leading and trailing spaces.
# This is a comment line.
The cost is $50.00 for the item.

vboxuser@Linux:~$
```

I used sed with the d command to delete specific lines. The line can be selected using its line number or a matching pattern.

39. Append or insert text before/after a pattern

```
vboxuser@Linux:~$ sed '/Error/i\Warning: Check the server connection.' hello
The quick brown fox jumps over the lazy dog.
Linux is a powerful operating system. Linux is open-source.
Warning: Check the server connection.
Error: Connection timed out at 10:42 AM.
User: admin, Access: granted, IP: 192.168.1.1
apple, banana, cherry, date, elderberry
123-456-7890
   This line has leading and trailing spaces.
# This is a comment line.
The cost is $50.00 for the item.

vboxuser@Linux:~$ sed '/Error/a\Action: Restart the service'. hello
The quick brown fox jumps over the lazy dog.
Linux is a powerful operating system. Linux is open-source.
Error: Connection timed out at 10:42 AM.
Action: Restart the service.
User: admin, Access: granted, IP: 192.168.1.1
apple, banana, cherry, date, elderberry
123-456-7890
   This line has leading and trailing spaces.
# This is a comment line.
The cost is $50.00 for the item.

vboxuser@Linux:~$
```

I used sed with i to insert text before a matching line and a to append text after a matching line.

40. Print only specific lines

```
vboxuser@Linux:~$ sed -n '2p' hello
Linux is a powerful operating system. Linux is open-source.
vboxuser@Linux:~$ sed -n '2,4p' hello
Linux is a powerful operating system. Linux is open-source.
Error: Connection timed out at 10:42 AM.
User: admin, Access: granted, IP: 192.168.1.1
vboxuser@Linux:~$
```

I used sed with the -n and p options to print only the required lines instead of displaying the complete file.

41. Copy a file from Linux to Windows

```
vboxuser@Linux:~$ cd ~
vboxuser@Linux:~$ echo "Hello from linux">task41
vboxuser@Linux:~$ ls
Desktop    Downloads  Pictures  Templates  book  cloud_training  dog   file_link  media  saam      snap      task41
Documents  Music      Public    Videos     cat   cow             file  hello      pen    saam.pub  students  test
vboxuser@Linux:~$
```

I used scp to securely copy a file from the Linux machine to a Windows machine over SSH.

42. Five SED use cases

```
MINGW64:/c/Users/Ayesha/SED

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/SED (master)
$ touch sedcases

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/SED (master)
$ vi sedcases

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/SED (master)
$ sed 's/unix/linux' sedcases
sed: -e expression #1, char 12: unterminated `s' command

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/SED (master)
$ sed 's/unix/linux/' sedcases
linux is great os. unix is opensource. unix is free os.
learn operating system.
linux linux which one you choose.
linux is easy to learn.unix is a multiuser os.Learn unix .unix is a powerful.

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/SED (master)
$
```

1. Replacing or substituting string: Sed command is mostly used to replace the text in a file

2. Replacing String on a Specific Line Number

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/SED (master)
$ sed '3 s/unix/linux/' sedcases
unix is great os. unix is opensource. unix is free os.
learn operating system.
linux linux which one you choose.
unix is easy to learn.unix is a multiuser os.Learn unix .unix is a powerful.
```

The above sed command replaces the string only on the third line.

3. To remove 1st character in every line:

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/SED (master)
$ sed 's/^.//' sedcases
nix is great os. unix is opensource. unix is free os.
earn operating system.
nix linux which one you choose.
nix is easy to learn.unix is a multiuser os.Learn unix .unix is a powerful.
```

.(dot) tries to match a single character. The ^ tries to match a pattern(any character) in the beginning of the line. Another way to write the same:

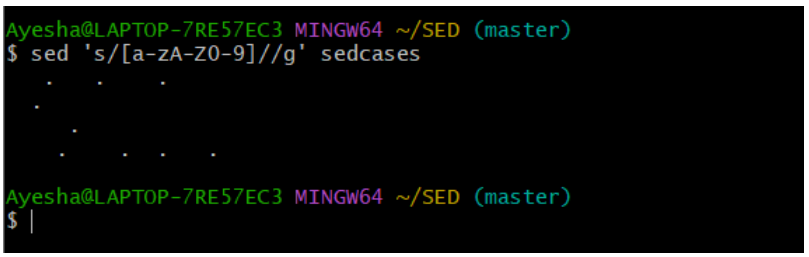
4. To remove last character of every line :

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/SED (master)
$ sed 's/ .$//' sedcases
unix is great os. unix is opensource. unix is free os.
learn operating system.
unix linux which one you choose.
unix is easy to learn.unix is a multiuser os.Learn unix .unix is a powerful.
```

The $ tries to match a pattern in the end of the line.

5.

To remove all alpha-numeric characters present in every line:

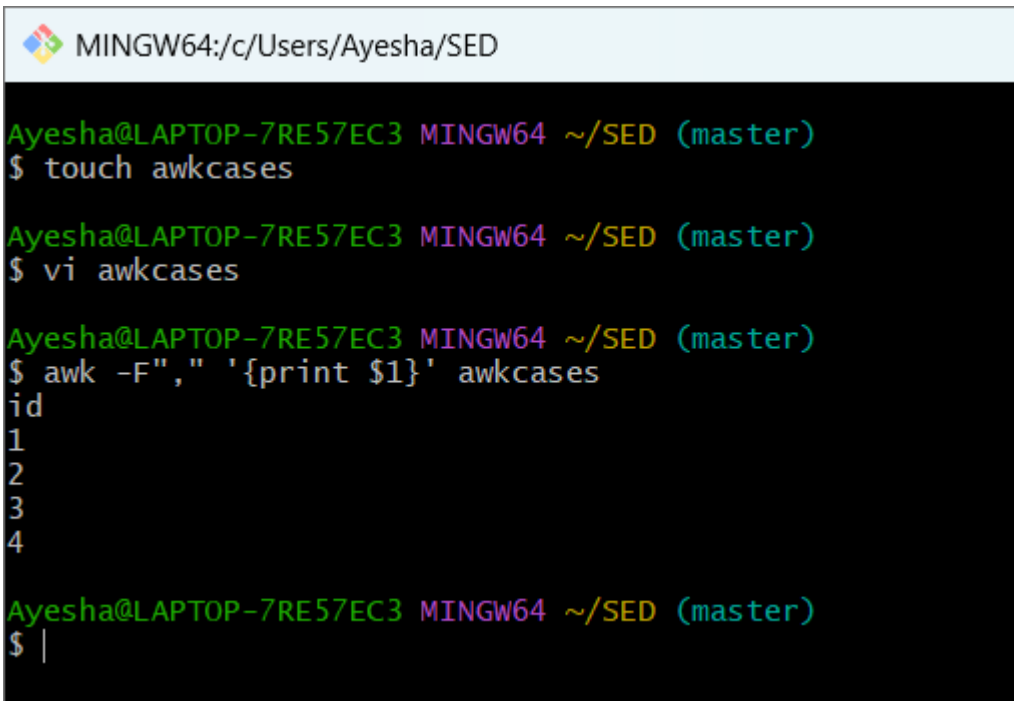


All alpha-numeric characters get removed.

AWK Use cases -

I used awk to perform calculations

1. Print First Column



I used AWK with the -F option to specify the field separator and $1 to extract the first column from the awkcases file. Since each line contains only one field, AWK printed the complete contents of the file.

2.Assign Variable

The -v option lets you assign a variable to be used within the awk script.

This is helpful for passing external values into the script.

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/SED (master)
$ awk -F"," -v var="Amount:" '{print var, $3}' awkcases
Amount: Amount
Amount: 100
Amount: 200
Amount: 150
Amount: 175
```

3. **Awk can perform complex data manipulation tasks.**

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/SED (master)
$ awk -F"," '{sum += $3} END {print sum}' awkcases
625

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/SED (master)
$
```

I used AWK to perform data manipulation by reading values from the third column and calculating their total. AWK processed each record, added the third-column values, and displayed the final sum.

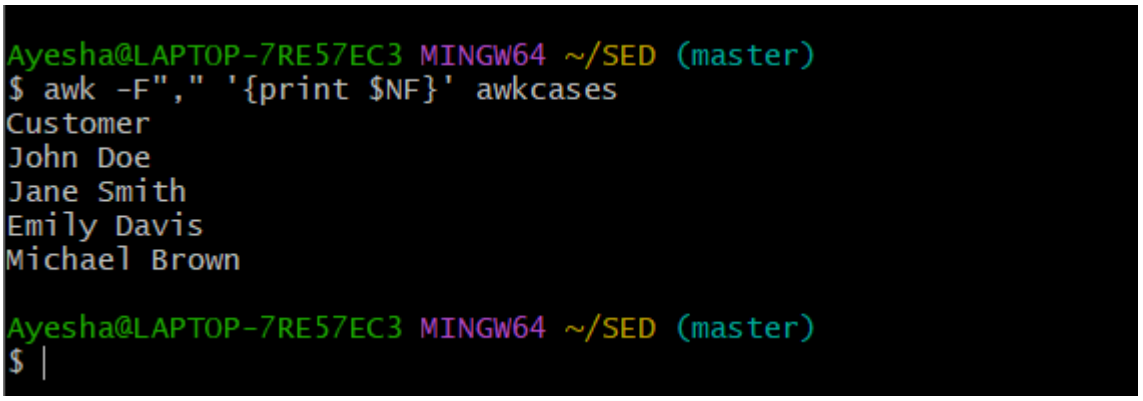
4. Print only USD transactions

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/SED (master)
$ awk -F"," '$4 == "USD" {print}' awkcases
1,2024-11-01,100,USD,Payment,John Doe
3,2024-11-03,150,USD,Purchase,Emily Davis

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/SED (master)
$
```

I used AWK to filter the file and print only the transactions where the Currency column contains USD. Since Currency is the fourth column, I used $4 == "USD" as the condition.

5. **Print the Last Column**



I used AWK with $NF to print the last column of the CSV file. NF represents the number of fields in the current record, so $NF refers to the last field regardless of how many columns the file contains.